tests can be applied in other ways. First of all, why are tests important and why do you need automated tests? Why can't you just test manually?

10.1. No tests? Disaster awaits

One time-honored approach to testing JavaScript applications is to test everything manually. Manual tests are important, but a testing strategy that uses only manual tests doesn't work. You've seen it many times before. It's not sufficient, it doesn't scale, and it's expensive because you need the real application and real humans to do all of it. Humans are *really* expensive, and you don't need to spend all their valuable time doing something the computer can do.

Manual testing is more difficult in the long run. If it's too difficult, you'll decide not to do the tests, and before you know it, *there are no tests*.

What happens when you don't write tests?

Imagine you've written all of the tracking code for Agtron's site. It tells him how many visitors his shop gets. Yesterday he had more than 100,000 visitors. You have no idea how he managed to get that much traffic to his website, but it's awesome nonetheless. Even better, your application and the accompanying tracking application, both written entirely in CoffeeScript, are performing nicely. No hiccups. As the sun sets before the weekend, you get a phone call from Agtron asking you to change the tracking program to add tracking of mouse movements for every user. Beer in hand on a Friday evening, you deftly implement a solution. On your way out the door, you go live to the production servers. Job done.

You awake in the night to the sound of your phone ringing. Something is wrong. The tracking program isn't tracking *any* users. It occurs to you to debug the client-side program on the live server to see if you can find the problem there. When you load it up, though, you remember that the code on the production server has been compressed to make it download faster. It's a big sea of unreadable code similar to this: